250 results for “(survey OR review) AND (wildfire OR “wild fire” OR “wildland fire” OR “wild land fire” OR “forest fire” OR bushfire) AND (propagation OR spread OR behavior OR dynamics OR trajectory) AND (prediction OR forecast OR simulation OR model OR management)”

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A Computationally Efficient Method for Updating Fuel Inputs for Wildfire Behavior Models Using Sentinel Imagery and Random Forest Classification

Amy L. DeCastro | Timothy W. Juliano | B. Kosović | H. Ebrahimian | J. Balch

Environmental Science, Computer Science · Remote Sensing · 17 March 2022

TLDR The need **for** up-to-date fuel maps available in real time to provide accurate **prediction** of **wildland fire spread** is demonstrated, **and** the methodology based on high-resolution satellite observations **and** machine learning that can accomplish this task is outlined. Expand

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Forest Fire reseArch

D. Viegas | Environmental Science, Engineering · 17 October 2014

Fire management systems materialize the integration of **fire** science **models** and decision support planning modules. Their operational usage often requires the concurrent execution of a large number of… Expand

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Coupled Dynamic Data-Driven Framework for Forest Fire Spread Prediction

C. Brun | A. Cortés | T. Margalef

Environmental Science, Computer Science · International Conference on Dynamic Data-Driven… · 5 November 2014

TLDR A Dynamic Data-driven methodology is used that is a calibration stage previous to the **forecast** where complementary **models**, data injection **and** intelligent systems are **w**orking in a symbiotic way to reduce the **forecast** errors at real time. Expand

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Spatiotemporal dynamics of forest fires in the context of climate change: a review.

Yuanjun Zhang · Hwee San Lim · Chengyu Hu · Rui Zhang

Environmental Science · Environmental science and pollution research… · 25 April 2024

Forest fires are sudden, destructive, hazardous, and challenging to manage and rescue, earning them a place on UNESCO's list of the world's eight major natural disasters. Currently, amid global… Expand

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Coupled Atmosphere-Wildland Fire Modelling

Jean Baptiste Filippi · F. Bosseur · +6 authors · J. Balbi

Environmental Science · 1 April 2009

Simulating the interaction between **fire** and atmosphere is critical to the estimation of the rate of **spread** of the **fire**. **Wildfire's** convection (i.e., entire plume) can modify the local meteorology… Expand

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A stochastic Forest Fire Model for future land cover scenarios assessment.

M. D'Andrea · P. Fiorucci · T. Holmes · Environmental Science · 13 October 2010

Land cover is affected by many factors includ- ing economic development, climate and natural disturbances such as **wildfires**. The ability to evaluate how **fire** regimes may alter future vegetation, and… Expand

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Uncertainty quantification of forecast error in coupled fire–atmosphere wildfire spread simulations: sensitivity to the spatial resolution

U. Ciri · M. M. Garimella · F. Bernardoni · R. L. Bennett · S. Leonardi

Environmental Science · International journal of wildland fire · 18 August 2021

A methodology to quantify uncertainty in **wildfire forecast** using coupled **fire**-atmosphere computational **models** is presented. In these **models**, an atmospheric solver is coupled with a **fire-spread**… Expand

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The FireWork air quality forecast system with near-real-time biomass burning emissions: Recent developments and

evaluation of performance for the 2015 North American wildfire season

R. Pavlovic · Jack Chen · +4 authors · S. Cousineau

Environmental Science · Journal of the Air and Waste Management… · 2 March 2016

TLDR A performance analysis of FireWork forecasts for the 2015 wildfire season shows that FireWork provides significant improvements to surface PM2.5 forecasts and valuable guidance to regional forecasters and first responders. Expand

77 · PDF · Taylor & Francis · Save · Cite

Modeling Wildfire Perimeter Evolution using Deep Neural Networks

Maxfield E. Green · K. Kaiser · Nat Shenton

Environmental Science, Computer Science · arXiv.org · 8 September 2020

TLDR A wildfire spreading model is proposed that predicts the evolution of the wildfire perimeter in 24 hour periods and is able to learn wildfirespreading dynamics from real historic data sets from a series of wildfires in the Western Sierra Nevada Mountains in California. Expand

6 · [PDF] · arXiv · Save · Cite

AEGIS: a wildfire prevention and management information system

K. Kalabokidis · A. Ager · M. Finney · Nikos Athanasis · Palaiologos Palaiologou · C. Vasilakos

Environmental Science · 12 October 2015

Abstract. We describe a Web-GIS wildfire prevention and management platform (AEGIS) developed as an integrated and easy-to-use decision support tool to manage wildland fire hazards in Greece (… Expand

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